

Unsupervised Categorization of Radio Pulsar Candidates

UE24CS352A Machine Learning, Mini-Project | Teammate 2: PES1UG24CS419 | Problem Statement 3, HTRU2 dataset

1. Scope and data

The supervised half of this project separates pulsars from noise. This half takes the next step and asks whether the pulsars themselves fall into distinct kinds. All models here are fitted on the 1,639 confirmed pulsars only (8 continuous features); the 16,259 non-pulsar rows are dropped, since including them would simply recover the supervised split. The features are summary statistics (mean, standard deviation, excess kurtosis, skewness) of the integrated pulse profile and of the DM-SNR curve. Because both k-means and the Self-Organizing Map compare points by Euclidean distance, and the raw columns differ by orders of magnitude, every feature is standardised to zero mean and unit variance first.

2. Method

K-means. Lloyd's algorithm with ten restarts, swept over $k = 2$ to 10 and scored by inertia and mean silhouette. **Self-Organizing Map.** A 5x5 Kohonen map written from scratch in NumPy. For each sample the Best Matching Unit is the node whose codebook vector is nearest in Euclidean distance; every node is then pulled toward the sample by $w \leftarrow w + \text{lr}(t) h(t) (x - w)$, where $h(t)$ is a Gaussian centred on the BMU and measured in grid coordinates. Measuring the neighbourhood on the grid rather than in input space is what makes the map topology preserving. Both the learning rate (from 0.5) and the radius (from 2.5) decay geometrically over 100 epochs. **PCA** is fitted after clustering and used for plotting only; clustering runs in the full 8-dimensional space, because projecting first would discard variance before either algorithm sees the data.

3. Choosing k

k	2	3	4	5	6
Inertia	7943	5989	4815	4091	3594
Silhouette	0.3676	0.3348	0.3425	0.3063	0.2857

Inertia falls fastest up to $k = 3$ and then flattens, which is the elbow. The silhouette disagrees and peaks at $k = 2$. We keep $k = 3$: it sits at the elbow, it matches the structure reported by Debesai et al., and the cluster profiles below show that the $k = 2$ solution merges two groups which differ physically. The disagreement is expected rather than troubling, since inertia measures compactness alone while the silhouette also rewards separation, and on data this continuous the coarser solution tends to win on separation.

4. The three groups

Cluster	n	mean_ip	std_ip	kurt_ip	skew_ip	mean_snr	std_snr	kurt_snr	skew_snr
0	739	71.57	42.39	2.04	6.82	38.65	57.91	2.48	8.04
1	679	28.62	33.19	5.01	29.19	77.09	66.12	1.06	1.52
2	221	93.18	43.37	1.01	2.87	3.44	21.99	8.90	101.43

Cluster 0 (739 candidates) is the typical population: a mid-range integrated profile with a moderately dispersed DM-SNR curve. **Cluster 1** (679) has a low profile mean, strong profile skew and the highest DM-SNR mean, so these are bright, sharply peaked candidates. **Cluster 2** (221) is the smallest and most distinctive: a high profile mean with a very weak DM-SNR curve and extreme DM-SNR skew.

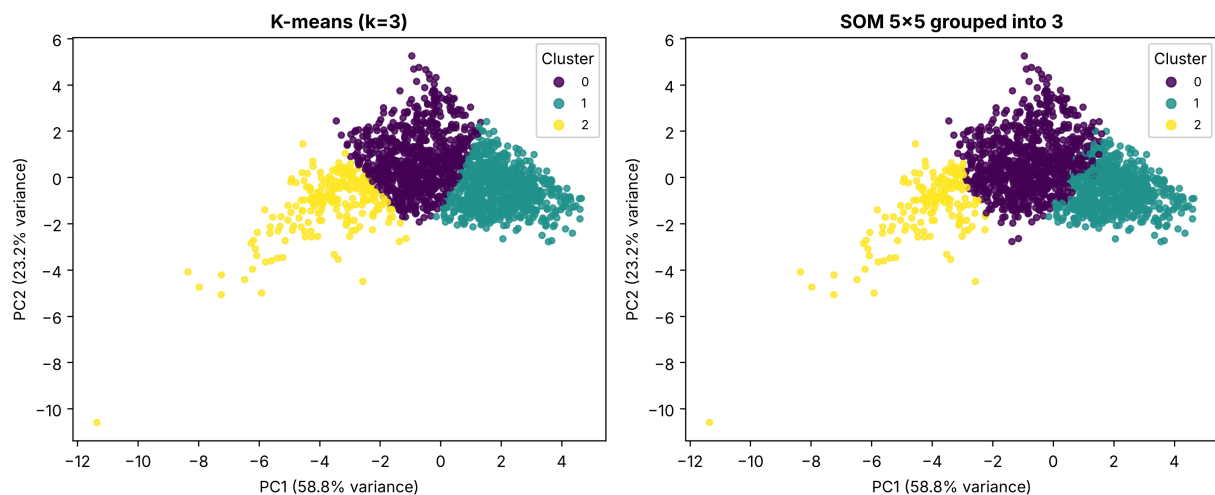


Figure 1. The same 1,639 pulsars under k-means (left) and the from-scratch SOM (right), plotted on the first two principal components. The two partitions are nearly identical.

5. Results

Method	Clusters	Silhouette
K-means	3	0.3348
SOM 5x5, codebook grouped into 3	3	0.3316
SOM raw BMU assignment	25	0.1715

A 5x5 map yields 25 micro-clusters, which is not comparable with k-means at $k = 3$. Following the two-level approach of Vesanto and Alhoniemi (2000), the 25 codebook vectors are clustered into three groups and each candidate inherits the group of its BMU; this is far cheaper than re-clustering all 1,639 samples. On that like-for-like basis the two methods differ by 0.0032 silhouette and agree on the group for **93.3%** of candidates. Two algorithms with entirely different mechanics converging on the same partition is the strongest evidence available here that the structure is a property of the data rather than of either method. The 25-node score is low by construction, not because the map failed: with 25 clusters over 1,639 points neighbouring nodes overlap heavily, which is exactly what the silhouette penalises.

The map itself trained cleanly. Quantization error fell from 2.028 after the first epoch to 0.861 after 100, and all 25 of 25 nodes are occupied, so there are no dead units and the grid is not oversized. The U-matrix shows its largest inter-node distances in one corner, which is where cluster 2 separates from the rest.

Two principal components retain 82.0% of the variance (PC1 58.8%, PC2 23.2%), enough for an honest 2D picture. PC1 sets the profile shape statistics (kurtosis +0.89, skewness +0.84) against the profile mean (-0.87) and DM-SNR kurtosis (-0.83), making it a "narrow, peaked profile" axis; PC2 is driven by dispersion (std_ip +0.64, std_snr +0.54).

6. Demo

A Streamlit application accepts the eight measurements for a single candidate and returns a pulsar verdict with its probability from the supervised GDA model, the k-means and SOM cluster assignments with the BMU grid coordinates, and a PCA scatter plot marking where the candidate falls among the known pulsars. Reusing the supervised model rather than reimplementing it keeps the two halves of the project consistent.

7. Conclusions and limitations

The pulsar population does carry internal structure, and three groups of 739, 679 and 221 candidates is a defensible description of it, independently recovered by two different algorithms. The separation is only moderate, however, and the limitation is the feature set rather than the algorithms. A silhouette near 0.33 says the groups are real but adjacent. HTRU2 provides summary statistics of the integrated profile and the DM-SNR curve, so frequency and timing information has already been averaged away; pulsar populations are best separated by period and dispersion measure, and neither survives into these eight columns. Adding them is the obvious next step. Three further caveats: there is no ground truth for the sub-types, so validation is internal only and confirming cluster 2 as a known class would require a catalogue cross-match such as the ATNF; $k = 3$ is a judgement call, and the silhouette alone would have chosen $k = 2$; and the SOM depends on its initialisation, so although the seed is fixed for reproducibility and the three-group structure survives reseeding, the map shifts.